

DAILY PRACTICE PROBLEMS (DPP-04) BASED ON EIGEN VALUE AND VECTORS

Q.1 The eigen values of the following matrix are :

$$\begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

(A) $3, 3+j5, 6-j$ (B) $-6+j5, 3+j, 3-j$

(C) $3+j, 3-j, 5+j$ (D) $3, -1+j3, -1-j3$

Q.2 If a square matrix of order 100 has exactly 15 distinct eigenvalues, then the degree of the minimum polynomial is

(A) at least 15 (B) at most 15

(C) always 15 (D) exactly 100

Q.3 The eigen vectors of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ are written in the form $\begin{bmatrix} 1 \\ a \end{bmatrix}$ and $\begin{bmatrix} 1 \\ b \end{bmatrix}$. What is $a + b$?

Q.4 Let A be a 4×4 matrix with eigen values $-5, -2, 1, 4$. Which of the following is an eigenvalue of

$\begin{bmatrix} A & I \\ I & A \end{bmatrix}$, where I is the 4×4 identity matrix ?



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Q.5 For the matrix $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$, ONE of the normalized Eigen vectors is given as

(A) $\begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$

(B) $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix}$

(C) $\begin{pmatrix} \frac{3}{\sqrt{10}} \\ \frac{-1}{\sqrt{10}} \end{pmatrix}$

(D) $\begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$

Q.6 The matrix $A = \begin{bmatrix} \frac{3}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2} \end{bmatrix}$ has three distinct Eigen values and one of its Eigen vectors is $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Which

one of the following can be another Eigen vector of A ?

(A) $\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

(B) $\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$



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Q.7 For the matrix $P = \begin{bmatrix} 3 & -2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, one of the eigen values is equal to -2 . Which of the following is an eigen vector?

(A) $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$

(B) $\begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

(D) $\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$

Q.8 The eigen values of a 2×2 matrix X are -2 and -3 . The eigen values of matrix $(X + I)^{-1}(X + 5I)$ are

(A) $-3, -4$

(B) $-1, -2$

(C) $-1, -3$

(D) $-2, -4$

Q.9 Given matrix $\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}$, the eigen vector is

(A) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

(B) $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$

(C) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

(D) $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

